

Charles Fucile

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Education

Georgia Institute of Technology

Aug 2024 – May 2028

Bachelor of Science in Aerospace Engineering – *John H. Martinson Honors Program*

GPA: 3.80

Relevant Coursework: Aerodynamics, Structural Analysis, Thermodynamics, Computational Fluid Dynamics, Jet & Rocket Propulsion, Orbital Mechanics, Classical Dynamics, Experimental Methods, Circuits & Electronics

Technical Experience

Aircraft Window Proof Fixture Modernization Intern

Aug 2026 – Dec 2026

NASA Armstrong Flight Research Center

- Redesigning and modernizing an aircraft optical window proof test fixture through conceptual design of a new fixture supporting larger window geometries and modern test requirements for 8 distinct aircraft on center.
- Performing CAD modeling, structural analysis, and feasibility trade studies to evaluate design options, review test requirements, and support fabrication planning for both fixture configurations.
- Took full ownership of design, drawing, and fabrication for 20+ additional hardware designs supporting airborne science mission operations at Armstrong.
- Shadowing aircraft operations engineers to gain direct exposure to real-time flight test planning, mission management activities, and the multidisciplinary dynamics of an active flight test center.

Computational Fluid Dynamics Research

Jan 2026 – Present

Georgia Institute of Technology

- Conducted CFD simulations in Ansys Fluent across four flow configurations: symmetric airfoil at varying angles of attack, internal flow through a converging-diverging nozzle, boundary layer development over a flat plate, and arterial flow with and without plaque occlusion.
- Generated structured and unstructured meshes, performed mesh refinement studies, and verified solution convergence to ensure numerical validity across all cases.
- Quantified pressure distributions, velocity fields, and force coefficients; compiled comparative tabulated analyses and technical presentations for each configuration.
- Conducted low-speed wind tunnel testing of symmetric airfoils across varying angles of attack and flow speeds, comparing experimental lift and drag results against computational predictions.

Leadership & Service

Student Mentor

Nov 2025 – Present

Aero Maker Space

- Mentored 200+ students across a high-traffic makerspace with 20+ machines, providing hands-on training and troubleshooting support across additive manufacturing (FDM, SLA, SLS), laser cutting, CNC milling, and mechanical assembly workflows.
- Performed equipment maintenance and repair on fabrication machinery, including full mechanical restoration of a band saw and ongoing repair of a table saw, diagnosing root causes and returning equipment to reliable operation.
- Managed 3D printer slicing, job queuing, and printer maintenance to sustain high-throughput rapid prototyping operations for student projects across the facility.
- Conducted independent training sessions on safe equipment operation for new users, developing clear instructional approaches for audiences with varying technical backgrounds.

Skills

Engineering: Aerodynamics, CFD Analysis, Structural Analysis, Test Operations, Mechanical Assembly, Prototyping

Software: Ansys Fluent, MATLAB, SolidWorks, Fusion 360, Creo

Manufacturing: Additive Manufacturing (FDM, SLA, SLS), Laser Cutting, Equipment Maintenance & Repair